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PLECTA: Overlap-aware instance reconstruction of dense strand networks from binary masks

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Abstract

Instance segmentation on a dense network of long, thin strands is difficult because it can fail in two opposite ways: a loose reconnect algorithm ends up fusing the wrong strands, and a strict one ends up fragmenting them, so that each strand is split into multiple pieces. A non-perfect binary mask, such as one from machine-learning segmentation, makes both failures harder to avoid, because it poses the same two ambiguities at every junction and every gap: a true crossing, where separate strands legitimately fuse into one connected component, cannot be told apart from an erroneous merge, and a true end cannot be told apart from a break left by a fading segmentation. Resolving either ambiguity takes the same judgement: which arm continues into which.

We present PLECTA, which makes that judgement from a binary strand-axis mask and nothing else. It cuts the skeleton into arms and junctions and solves every junction, of any degree, as a single exact maximum-weight *partial* matching in which leaving an arm unmatched carries an explicit price. A strand is therefore allowed to end, instead of having a continuation invented for it. Crossing pixels are emitted to every strand that claims them, so the output is a set of overlapping layers rather than the hard partition most instance methods return. The evidence is purely directional (tangent and curvature, read over tens of pixels either side of the discontinuity), which makes the working condition plain: the method applies where direction survives a crossing, as it does for nanotube bundles, biological filaments such as actin and collagen, and synthetic fibres, and not for freely flexible chains.

On 128 synthetic scenes from unseen seeds, PLECTA reaches mean pairwise $\mathrm{F}_1 = 0.854$ [0.845, 0.864], falling from 0.911 to 0.775 as coverage rises from 20 to 60 %. It leads two released implementations, run from their own sources on identical masks, at every density. The comparison was repeated on a competitor's own generator, GraFT's, vendored unmodified at its published settings, over 40 scenes. The ordering held at every density stratum, at F_1 0.916 against 0.475 and 0.349.

A secondary depth-resolution stage then reads the greyscale to decide which strand lies on top at each crossing, and stacks the network into layers, an ordering otherwise obtained by destructive sectioning. It is correct on 0.885–0.961 of the crossings it commits to, against a chance level of 0.5, because it abstains where the image does not separate the two by more than the local noise. It abstains at 8–18 % of crossings, and counting every one of those as an error it is correct on 0.556–0.815 of all reference crossings. The method uses no random number generator and its configuration is published, so a mask yields the same strands on every run. We demonstrate the whole workflow on 3 annotated carbon-nanotube SEM fields, and state plainly what that evidence does not support.

Keywords: strand networks; instance segmentation; graph matching; geometric reconstruction; carbon nanotubes; scanning electron microscopy.

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1 Introduction

Many materials are composed of long, thin strands laid over one another, some tangled at random and some deliberately twisted: sheets of carbon nanotubes, collagen in connective tissue, paper, felt, nonwoven textiles, fibre-reinforced composites, aramid fibre ropes and synthetic fibre mats. The effective properties of such a material (tensile and flexural stiffness, electrical and thermal conductivity, fracture, tear and impact resistance) are governed as much by the architecture of the network as by the constituents themselves, and often more so: the stiffness and strength of a nanotube assembly are bounded by load transfer between tubes rather than by the very high modulus of any one of them [1–6]. The governing descriptors are therefore geometric: the length of each strand, its thickness, its orientation, its local curvature, and its connectivity to the strands around it. The same descriptors are the measurement target well outside materials science. A DNA fibre spreading assay reads the length ratio between two labelled replication tracts along an individual fibre [7], and time-resolved imaging of the actin cytoskeleton follows how its filaments reorganise inside a living cell [8].

A micrograph shows that network directly. Turning the picture into those numbers is the hard part. Routine image analysis measures the network in bulk (what fraction of the frame is covered, which way the texture points [9, 10]) and stops there. To measure a distribution of strand lengths, or to ask how often one strand crosses another, something more is needed: every strand must be recovered as a separate object, addressable from one end to the other.

That is an instance-segmentation problem, and it has an awkward geometry. The objects are long, thin and open rather than compact and closed; they touch, cross and hide one another; and what tells two of them apart is not shape or size but which way each was heading. The consequence is a pair of failures that pull in opposite directions. Where two strands cross, their pixels join into a single connected blob, the same blob a genuine merge error would leave, and a method that trusts connectivity reports one object where there are two. Where the image fades or the segmentation drops out, one strand arrives in several disconnected pieces, the same gap a genuine end would leave, and the same method reports several objects where there is one. Both situations are repaired by answering one question: at this junction, which arm continues into which?

Existing tools answer part of it. Semantic segmentation is mature, since U-Net and its self-configuring descendants reliably separate strand foreground from background [11, 12], but a foreground mask carries no identity, so it inherits both failures rather than resolving them. General-purpose instance methods assume the objects are compact: bounded blobs, centre-directed flows, star-convex outlines, none of which describes a long open curve that may overlap another [13–15]. Ridge and vesselness filters sharpen curvilinear signal but assign nothing across a junction [16–18], and the established fibre-analysis packages return network statistics rather than resolved objects [19–23]. Where the imaging is volumetric the problem is different in kind, because strands that appear to meet occupy disjoint volumes: in X-ray tomography of fibrous solids and of twisted ropes, individual fibres are separated by seeded watershed and morphological reasoning on the reconstructed volume [6, 24]. A projection has no such recourse, since two strands that cross genuinely share the same pixels. This paper addresses the step those leave open: given a binary strand-axis mask, recover the individual strands through the crossings and the gaps.

Recovering strands through crossings is well-trodden ground. Existing methods differ in two ways that matter: what evidence they read, and whether a pairing once made can later be undone. On both counts the field clusters.

The largest cluster follows one recipe. These methods skeletonise the mask, cut the crossing out, and pair the arms left behind by how nearly each continues the direction of another. Each pairing is accepted as soon as it is scored. SIFNE reads its tip directions from the excised stubs and recovers dense microtubule networks from the mask alone [25]. Wegmayr et al. pair skeleton branches by lowest mutual angle [26]. GraFT covers a Frangi-filtered skeleton greedily, longest path first [8]. This is cheap, and where crossings are rare it is enough. Its weakness is structural: once an arm is paired, a later arm that contradicts the choice cannot reopen it.

Two ways out of that have been tried. The first is to bring in more evidence than a mask carries. Wegmayr et al.'s stronger arm drops the angle rule and learns pixel embeddings on patches around each junction; it reads the greyscale image, and it must be trained on crops in which every fibre is already labelled individually. Even then it recovers tangled fibres at about half the precision of isolated ones, in data where two thirds of fibres cross nothing at all. Methods that pair ends with an orientation network, detect fibres with a region-based detector, or trace them recurrently [27–29] need the same labelled training data. Reading a mask and nothing else avoids that cost, which is the trade this paper takes. DNAi keeps to the mask and simply looks harder, enumerating every pairing a junction allows from its endpoint angles; it does so exhaustively at degree three and four, and not at all above [7].

The second way out is to stop deciding one pairing at a time. Basu et al. and DeFiNe both weigh a whole neighbourhood at once, which is what a busy node needs. Basu et al. delete every branch point from a spanning tree over centreline points, then reassemble the pieces with an integer program that prices each join by how sharply it bends, repeating until the set of pieces settles; a join that a later fit contradicts can therefore still be undone [30]. DeFiNe covers an already-built weighted graph with the smoothest paths it can find, and lets two paths share an edge, so a crossing pixel can belong to both strands instead of being awarded to one [31]. Neither, though, starts from an axis mask: one wants intensities, the other a graph some earlier stage has already produced. Both also reassemble at most three pieces at a time, which limits how much of a shattered neighbourhood a single round can repair. Deciding broadly and starting from a mask have so far been alternatives.

One method has them both, and it comes closest to the position taken here. De et al. reason globally and do start from a mask. Branches and junction contacts become a graph, and labels spread across it so that a decision at one node constrains its neighbours rather than resting on that node's own angles [32]. The labels have to start somewhere, however. In their images they start at anatomical roots, the stained cell bodies of neurons or the optic disc of a retina, and the method returns as many strands as there are roots. A tangle of open-ended strands offers no such starting point, and how many strands it holds is precisely the unknown. Reasoning globally from a mask alone, with no seed and no strand count supplied in advance, is where this paper begins. Four of these methods are measured against it directly in Section 3.3, on identical masks and through one scorer.

Skeleton graphs, junction excision, continuation angles, distance gates, combinatorial optimisation, iterative refitting and overlapping output are therefore all prior art, and none is claimed here. What is new is narrower, and it is a combination: from a binary mask alone, PLECTA solves each junction of *any* degree as one exact maximum-weight *partial* matching in which leaving an arm unmatched is an explicit option at a fixed price rather than a parity leftover, alternates that with globally gated gap matching, re-fits its tangent and curvature frames along the chains as they assemble, and emits overlapping strands. Each element exists somewhere above; no method there combines them. Degree-general exact matching with a priced refusal is what lets a strand genuinely end rather than have a continuation invented for it. The closest single prior method is that of Liu et al., which likewise reads a mask, reconnects across gaps and permits overlapping output, but pairs termini greedily and depends on a trained orientation network [27]. That claim carries a condition, worth stating as scope rather than leaving to be inferred. The only evidence PLECTA has at a crossing is directional: the angle between two outward tangents, the turns onto the chord between them, and the continuity of signed curvature, each read over tens of pixels of arclength. The working condition is therefore a property of the image: direction must persist across the junction at that scale, or the continuation is not decidable from geometry at all. Bending stiffness is what most directly produces such persistence, which is why the semiflexible regime is the method's home ground: nanotube bundles sit there, and so do the actin, collagen and synthetic-fibre networks the methods above were built for. Stiffness contributes to the condition rather than uniquely causing it, since tension, adhesion, confinement or bundling can hold a direction just as straight. Where the condition fails, as for a freely flexible chain whose tangent decorrelates within the junction, no continuation is more plausible than another, and no amount of matching recovers what the image no longer distinguishes.

We introduce PLECTA, after the Latin *plecta*, a braid. It decomposes a skeleton into arms, endpoint stubs and merged junction nodes, alternates exact per-node continuation matching with globally gated gap matching while tangent and curvature estimates expand from local arms to assembled chains, and returns projected 2-D strand layers in which crossing pixels may be shared. Every parameter is fixed in a published configuration and the implementation draws on no random number generator, so a given mask yields the same strands on every run; that is the only sense in which it is called deterministic here.

Reaching real micrographs takes one further component, and the workflow is deliberately divided in two. An upstream route turns a raw SEM micrograph into a binary strand-axis mask by a synthetic-data-driven CycleGAN/U-Net whose training pairs are synthesised rather than annotated [33, 34]; PLECTA is everything downstream of that mask. The division is what makes the grouping measurable on its own, against references whose identities are known exactly, and it keeps width rendering from moving an identity score. The closest complete workflow is that of Peters et al., who join a U-Net-like segmentation trained on over a thousand annotated SEM micrographs to algorithms that trace fibres through intersections, and validate against human evaluators on fibre number, length and width [35]. That work and this one divide the task at the same joint and differ in what the instance step is asked to prove: their tracer stands in for human counting and measurement, while PLECTA is isolated behind a binary mask and scored on identity through crossings against overlap-aware references in which every pairing is known. We evaluate PLECTA on 128 procedurally generated scenes spanning 20 to 60 % areal coverage. A further 50 scenes are rendered twice, once from a clean axis and once from a degraded one, so that what degradation costs is measured on unchanged geometry. Both stages are then demonstrated together on 3 annotated CNT-sheet SEM fields.

Beyond the plane, a crossing carries a second question the grouping does not answer: which strand lies on top. For nanostructured networks that is normally obtained destructively, by focused-ion-beam sectioning and tomographic reconstruction [36], because a micrograph's intensities record surface topography rather than height. PLECTA answers it non-destructively from the greyscale already in hand. The machinery is deliberately conventional: monocular depth ordering has long derived pairwise depth from local occlusion evidence, assembled those relations into a depth-order graph and corrected the contradictions among them [37, 38], and single-view hair modelling resolves the layering of intersecting strands in much the same spirit [39]. What differs is the evidence available. Those methods read the shape of the outlines where two objects overlap, which a bundle a few pixels across in a noisy micrograph does not present reliably. What separates the two strands here is instead the continuity of each one's own brightness: the upper strand keeps the appearance it has on either side of the crossing, while the lower strand's is overwritten by the strand above it. The stage abstains where that difference is no larger than the local image noise. To our knowledge this is the first depth decomposition of a strand network from a single micrograph; it is reported as a demonstration with a measured accuracy, not as a validated three-dimensional reconstruction.

2 Materials and methods

2.1 Scope and workflow

The input is a binary strand-axis mask $M \in \{0,1\}^{H\times W}$, whose foreground $F = \{x : M(x) = 1\}$ is one pixel wide and traces strand centrelines rather than strand widths. From it PLECTA reconstructs an overlap-aware collection $\{I_1, \ldots, I_N\}$ with $I_n \subseteq F$, so that every emitted pixel is one the mask already carried, and a crossing pixel may belong to several strands at once. The evaluated grouping stage uses only M; a greyscale SEM image is an optional downstream input for width and brightness measurement and cannot revise the fixed grouping.

Figure 1 places the method. Its five geometric steps read mask geometry alone, and the two matching steps are re-made over eight rounds against frames re-fitted along the chains assembled so far. Where the mask itself comes from is a separate question, taken up in Section 3.5.

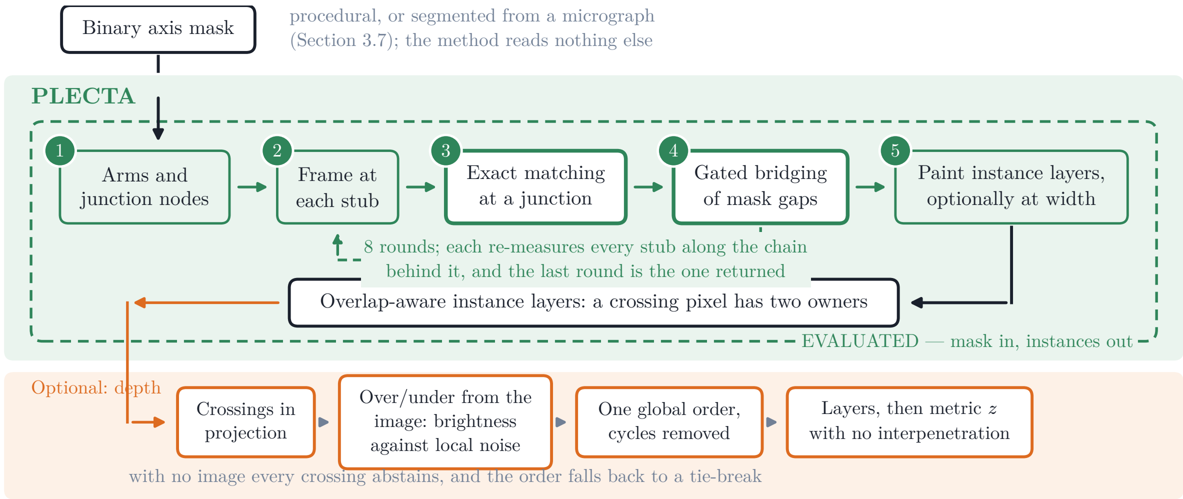


Figure 1. Where PLECTA sits. A single binary axis mask is the method's only input, and every result in this paper measures the dashed green box. Steps 3 and 4 are the only decisions it makes and are drawn heavier for it; the rest is bookkeeping. The optional depth stage adds a third dimension without revising any identity. Colours: green, a PLECTA geometric step; orange, a dependency on the greyscale image; dark, a data object.

What makes the task hard is a single crossing: the mask holds one connected blob from which several arms leave, and nothing local states which arm continues into which. The output is a set of layers rather than a partition, so a pixel inside that blob is counted in both strands that claim it.

2.2 Synthetic data and evaluation split

Because the scenes are generated rather than annotated, the answer is known exactly. For every strand k the generator records its ordered centreline $\mathcal{C}_k$ and the pixels it was drawn onto, its support Ω_k. The supports are kept one per strand instead of being merged, so a crossing pixel belongs to both strands that pass through it and the reference is overlap-aware by construction.

Each scene is written out four ways: the exact one-pixel axis; that axis degraded with small gaps and raster irregularities; the strands at full width, merged into a single foreground; and an SEM-like greyscale image, in which each strand carries its own brightness and a slow variation along its length, blurred by one point-spread function over a low-frequency background with grain. A depth-resolved variant of the generator supplies the missing structure for the depth stage of Section 2.5.2. It gives every strand a height and a radius, refuses to let two interpenetrate, and renders the stack from the top down, so the image carries occlusion, defocus that grows with distance from a randomised focal plane, and width broadening. Those cues are randomised independently of one another, so no single rule such as brighter-is-higher recovers the order.

Scenes are labelled throughout by areal coverage, the fraction of the frame the full-width strands cover rather than the fraction the thin axis occupies. That label reports difficulty without setting it: drawing the same strands thicker raises coverage and leaves the score exactly where it was, since neither PLECTA nor the metric ever sees how wide a strand was drawn. What makes a scene hard is the amount of centreline packed into the frame, which coverage happens to follow under this generator, and Section 3.1 separates the two.

Every parameter was fixed on a separate development set and never on the scenes reported here; the 128 evaluation scenes, 32 at each of 20, 30, 40 and 60 % coverage, use seeds that appear nowhere in that development set. Supplementary Section S6 gives the seed blocks, the verifier that checks their disjointness, and the locked set that remains unevaluated.

A separate controlled paired set of 50 scenes, ten at each of 20, 30, 40, 50, and 60% coverage, was scored once from each geometry's clean axis and once from its degraded mask, after the configuration was fixed.

2.3 CNT SEM data and mask sources

Three manually annotated CNT-sheet SEM fields were available for exploratory analysis, each 1536×1024 px: B58_100 (307 visible CNT bundles), B58_110 (430) and B58_300 (325). The annotated strand is the visible bundle, not the individual nanotube, which SEM does not resolve at this magnification. The nominal horizontal field width is 1.66 µm across the long, 1536-px axis of the stored images, which is 1.08 $\mathrm{nm\,px^{-1}}$. Each annotation is one human interpretation of projected, locally ambiguous crossings.

The reference annotations were drawn with an in-house interactive instance-segmentation tool for greyscale SEM of thin, crossing strands, whose data model is per-instance masks rather than a flat label image, so a pixel may belong to several strands at once; that is what makes the reference overlap-aware and comparable with PLECTA's output. The file the tool saves is the file the evaluation reads, with no export or conversion step in between. The annotator paints each visible bundle as its own strand, and the tool assists without deciding: strokes may snap onto the local intensity ridge, with polarity calibrated from the existing annotation, and may be re-rendered through a smoothing-spline fit whose centreline is shifted onto the measured intensity crest only where that crest is consistent; widths come from perpendicular full-width-at-half-maximum profiles referenced to the local minima either side of the peak and reported as twice the nearer half-width, so a crossing strand fusing into the profile does not bias the measurement wide. Candidate gap bridges, collinear endpoint pairs of different strands, are suggested but never applied automatically; each is accepted or rejected by the annotator, so identity through every crossing is a human decision, machine-assisted but not machine-made.

The protocol's limits are stated rather than implied. The reference annotations are annotator A's, and A is an author of this paper who also configured PLECTA. There was no blinding, so the reference may encode the method's own reading of an ambiguous crossing, and no real-data difference smaller than the variation between two annotators should be read as real. These are manual reference annotations, not ground truth. Annotator B has annotated 3 of these fields independently, and Section 4.1 reports what that measurement shows: the two annotators do not implement one strand definition, so every real-data score in this paper is reported against a named annotator and the two are never averaged. Scores throughout are computed on centreline fragments, not on strand widths.

Two inputs were derived per field. The first is the skeleton of annotator A's strand annotation, an oracle-like condition that asks what the grouping can do when the axis evidence is curated. The second is an automatic mask from the upstream CycleGAN/nnU-Net segmentation of Section 3.5 [12, 33, 34], which asks what an automatic mask allows it to do. That segmentation is a separate contribution with its own evaluation; all that matters here is that its output is a binary mask produced without seeing any annotation.

Mask quality was compared after skeletonisation with the symmetric 2-px tolerance used throughout, which avoids rewarding mask thickness, and reconstruction scores were computed within each mask's own visible common-fragment population.

The fields were acquired on a Thermo Fisher Helios 5 UX DualBeam with an Elstar field-emission column in immersion mode, imaged with the through-lens detector at 2.00 kV and 400 pA and a working distance of 4.3 mm. Each frame is an integration of 35 passes at 100 ns dwell with drift correction enabled.

2.4 PLECTA reconstruction

2.4.1 Skeleton graph and stub frames

PLECTA skeletonises the mask and prunes the short spurs skeletonisation leaves behind. Skeleton pixels where three or more paths meet form junction clusters, and clusters close enough together are merged into a single *node*: a crossing seen at this resolution is one blob, not a point, and neighbouring crossings often touch. What remains between nodes is an *arm*, an unambiguous run of strand, carrying a *stub* at each end.

Each stub is given a frame: its tip, the direction the strand is heading as it leaves, and how it is curving. Both are fitted over a short run of arclength back from the tip, and a stub too short to yield a reliable direction is marked as such rather than guessed at. That support widens once arms are linked into chains, so a stub with no direction of its own borrows one from the chain it has joined, which is why the method iterates rather than deciding once.

2.4.2 The pairing cost

Linking two stubs is priced by asking three geometric questions (Figure 2). Do the two directions oppose one another, as they would if one strand simply continued into the other? Does each direction point at the other's tip, rather than running parallel past it? And does the curvature continue, or would the join require a kink? For a gap the distance to be spanned is charged as well; at a junction there is no gap to pay for. The cost of linking stubs a and b is therefore

$$C_{ab} = w_d\,\theta + w_t\,q(d)\,(\varphi_a + \varphi_b) + w_\ell\,\frac{d}{\ell_0} + 30\,w_\kappa\,|\kappa_a + \kappa_b|, \qquad q(d) = \min\Big(1, \tfrac{d}{d_0}\Big), \tag{1}$$

with d the tip separation, θ the reversal angle between the two outward tangents, and φ_a, φ_b the turns from each tangent onto the tip-to-tip chord, measured at their own vertices; junction links set $w_\ell = 0$.

Three properties of that expression are deliberate. The reversal angle ignores the chord, which keeps it meaningful when the tips are three pixels apart and the chord direction is quantisation noise. The chord turns do see lateral offset, so two parallel arms that never meet are penalised. And because it is the chord terms that degrade over short separations, the fade $q(d)$ withdraws them as the tips approach rather than letting an unstable angle dominate. The weights, the scales ℓ_0 and d_0, and the constant placing curvature on the scale of the angular terms are all fixed in the released configuration and are listed in Supplementary Section S1.

2.4.3 Candidate gates and exact matching

Leaving a stub unmatched is priced rather than forbidden, and that price does double duty: it also fixes what may be offered at all, since an edge costing more than the two prices it would save could never be chosen. Exact maximum-weight partial matching then picks, over disjoint pairs,

$$\max \sum_{(i,j)\in\mathcal{M}} (p_i + p_j - C_{ij}), \qquad \text{admissible iff } C_{ij} < p_i + p_j, \tag{2}$$

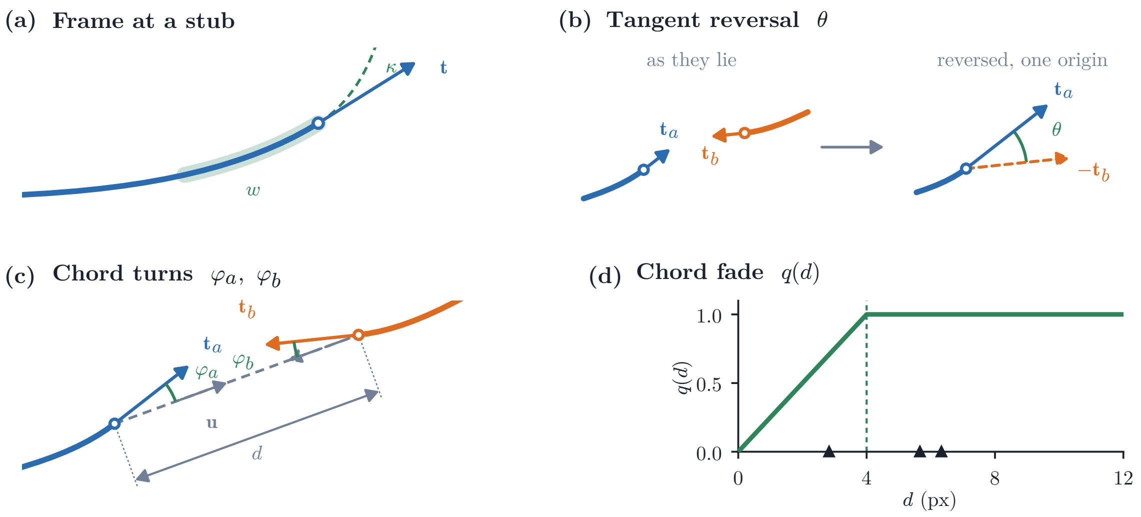


Figure 2. The stub frame and the terms of the pairing cost. **(a)** The least-squares arclength fit over window w, read outward from the tip at $s = 0$, giving outward tangent $\mathbf{t}$ and signed curvature κ; quadratic when the span allows, linear otherwise. **(b)** θ, in the two steps its definition takes: the outward tangents where they lie, then the same pair with $\mathbf{t}_b$ reversed and both read from one origin. A perfect continuation makes them antiparallel, so $\theta = 0$. **(c)** The chord turns φ_a, φ_b onto the chord $\mathbf{u}$ joining the tips. **(d)** The fade $q(d)$ against tip separation; triangles mark the six real separations at the crossing of Figure 3.

which is the same thing as minimising the total link cost plus a price p_i for every stub left free. Junction and gap links carry different prices, and a gap link must additionally clear limits on how far it may reach and how far either direction may turn, the tighter set, because a gap asserts a continuation across evidence that is *missing* rather than merely ambiguous. Supplementary Section S1 lists the prices, the limits, and the schedule below.

The method does not decide once. It runs the two matchings over several rounds, re-fitting every frame along the chains assembled so far before each. Early rounds are deliberately conservative: prices are scaled down, which admits fewer edges, and the gap angle limits are narrowed, so only pairings that survive the stricter test are committed while the frames are least reliable. The constraints relax towards their configured values as the chains sharpen the frames, and only the final round runs at the configured parameters. That round is the one returned.

Every stub is therefore either paired or pays the unmatched price, and there is no third possibility. That priced option is what lets a strand genuinely end, and what stops a node with an odd number of arms from inventing a continuation for the one left over. No node is classified: a node of any degree is solved by the same matching, and an arm ends because leaving it unmatched won the objective, not because the node was recognised as a particular kind.

It also removes a parameter rather than adding one. Admissibility in Equation 2 is not a threshold chosen alongside the price: an edge costing $C_{ij} \geq p_i + p_j$ contributes nothing positive to the objective, so a maximum-weight *partial* matching could not select it under any setting, and declining to offer it removes only edges that could never have been chosen. One number therefore states both what a refusal is worth and what may be considered, where a method that gates separately must set a cost threshold and a price and keep them consistent.

Solving a junction as one matching rather than one edge at a time is the distinction claimed in Section 1, and Figure 3 shows what it buys on real nodes, alongside the other two decisions the linker makes. A locally attractive edge can be declined, as at the three-arm node of Figure 3b. Joint and sequential solutions coincide at plain crossings and diverge only as nodes grow dense: they never disagree at nodes of degree two to four, disagree at roughly a fifth of nodes with five to eight stubs, and at two thirds of nodes with nine or more, where the joint solution is the

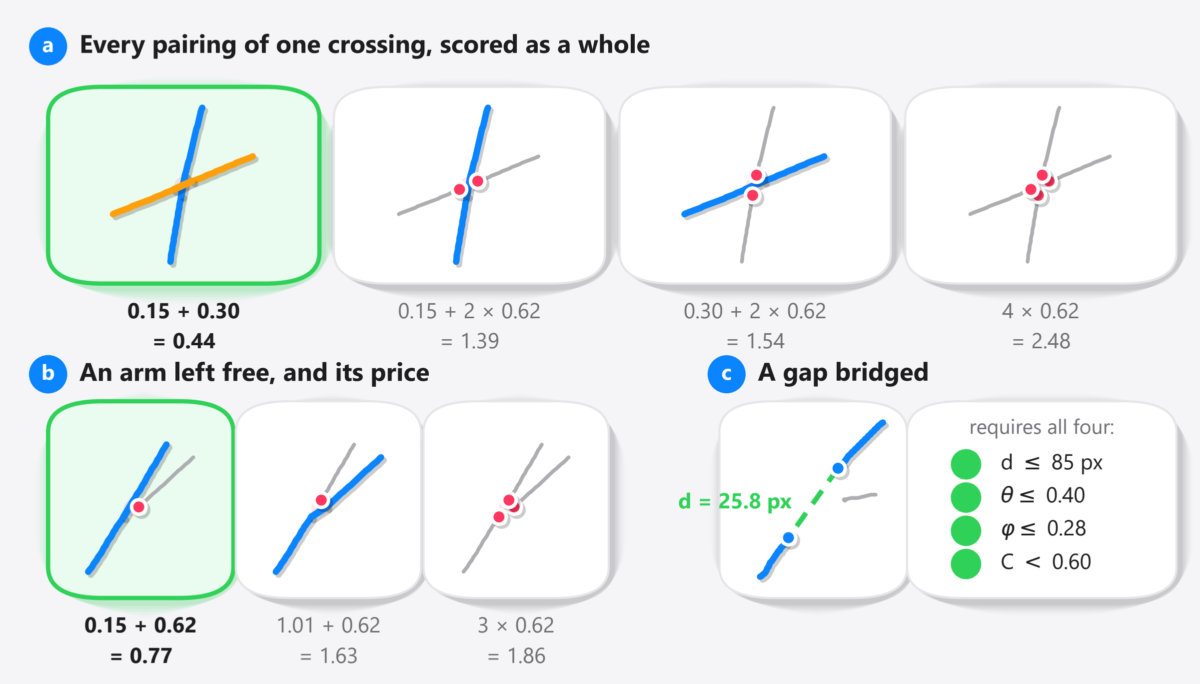


Figure 3. Every decision the linker makes, on the nodes it made them at. Every arm is the skeleton the numbers were measured from, so every angle shown is the one the cost was computed on. **(a)** Every pairing the admissibility gate allows at a four-arm crossing, each scored as a whole: its edge costs plus the price $p_i = 0.62$ of each arm it leaves free. The green card is the one returned; two further pairs exceed $p_i + p_j = 1.24$ and are never offered. **(b)** A three-arm node where an admissible edge is declined on the total rather than on its own cost. **(c)** A gap link, which must clear all four gates at once.

cheaper one; Supplementary Section S1 gives the counts. That comparison holds one node's costs fixed. Run inside the rounds, where each matching sets the frames the next is scored against, a cheap early choice propagates and the difference is no longer confined to dense nodes.

Two measurements test that claim rather than assert it. The greedy continuation baseline of Section 3.2 reads the *same* masks and the same local geometry and differs only in pairing greedily instead of jointly, so the +0.145 F_1 between them isolates the formulation rather than the evidence. The component ablations reported alongside it then ask the complementary question, whether the rest of the combination is load-bearing or decorative, and find that removing the junction chord-turn term costs -0.1858 and removing gap matching -0.1394. Neither measurement alone would do: the first says the matching is worth its complexity, the second that the terms it matches on are not redundant.

2.4.4 Strand output

Connected components of accepted arm links define strands. PLECTA paints every arm pixel and every junction cluster touched by a chain into that strand, so intersecting strands share crossing pixels. An isolated one-arm component shorter than 6 px is omitted. Accepted gap links determine identity but are not painted into the fixed centreline output. All effective parameters, including values previously inherited as code defaults, are materialised in the fixed JSON configuration.

2.5 Image-informed reconstruction

The grouping above reads the mask alone and is what this paper evaluates. Where a greyscale image of the same field exists, three further stages can read it: two give each strand a width, and one orders the strands at their crossings. All three run after the grouping is fixed and none can move a pixel from one strand to another, so no identity score in this paper depends on any of them. All are governed by limits recorded in the fixed configuration, and none is specific to SEM: they take whatever image accompanies the mask, and SEM is simply the case demonstrated here.

2.5.1 Width and ribbon rendering

Perpendicular intensity profiles, sampled along each ordered chain away from junctions, attach an apparent width, a brightness, an uncertainty and a quality flag to an existing strand without altering its arm membership. Ribbon rendering then interpolates accepted gaps, resamples and smooths the centreline, and draws the chain at its fitted width, excluding junction blobs. Neither silhouette absorption, which would union rendered strands, nor any other step merges two strands after the fact, and none contributed to the held-out result.

2.5.2 Depth decomposition

A projected crossing has an owner in each of two strands but no order between them. This stage supplies one.

Evidence at one crossing (Figure 4). Where two strand centrelines intersect, a short arc of each is taken through the crossing (the shared *core*, a fixed length, the same at every crossing), together with the *flanks* on either side of it, kept clear of the other strand's stroke so that they measure one strand and not the union. One channel is read on both: median intensity. The physical argument is occlusion. Whichever strand lies on top presents the same surface through the crossing as it does beside it, so its core matches its own flanks in brightness; the lower strand is overwritten there, so its own continuity breaks. The strand whose flank brightness the core matches better is called upper, and by how much, divided by the larger of the two strands' brightness separation and the noise measured at that crossing, is the score s that decides it. Where the two strands genuinely look alike, or differ by less than that noise, $|s|$ falls below a fixed threshold and the crossing *abstains* rather than being decided by noise. Dividing by the noise, rather than by the separation alone, is what makes a confident score mean *separated by more than the noise* rather than merely separated.

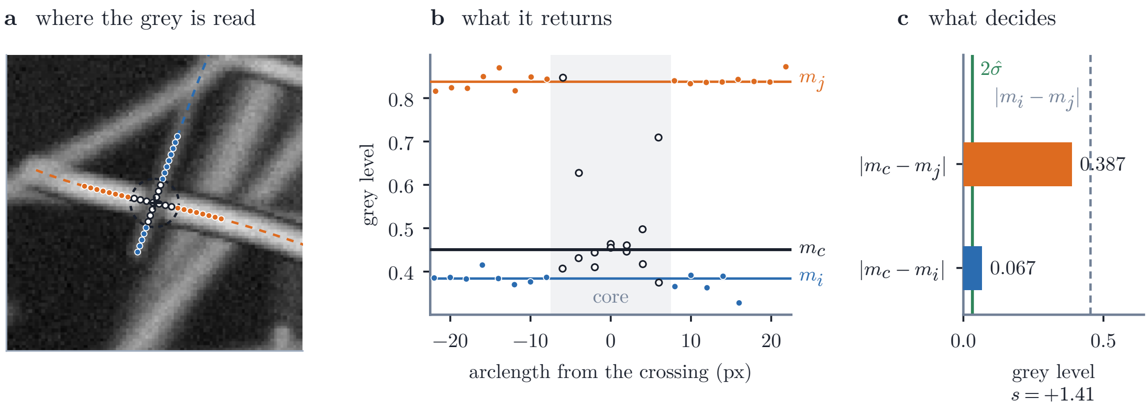


Figure 4. What the depth stage measures, at one crossing where two strands meet at 83.8°, chosen as a legible example rather than at random. Every position and grey level was read back from the stage itself. **(a)** The core arc shared by the two strands, and each strand's own flanks, which the window keeps clear of the other's stroke; open dots are core samples, filled ones flanks. **(b)** The same samples against arclength: the pooled core median m_c sits by strand i's flank level and far from j's, which is the occlusion. **(c)** The two distances the score compares, over the larger of the strands' contrast and twice the flank noise. That ratio is the score s of Supplementary Section S2, whose sign names the upper strand and whose size weights the crossing below.

One order for the whole field. Local decisions need not be mutually consistent: they can assert that A is above B, B above C and C above A, which no assignment of heights satisfies. Each decided crossing therefore contributes a directed edge weighted by $|s|$, the strength of its own evidence (Figure 5), and within each connected component of the crossing graph one total order over its strands is derived so as to maximise the total satisfied weight, which is the same as reversing the lightest set of relations that leaves no contradiction behind. That is the minimum-weight feedback arc set objective, and it is attained exactly only for components small enough to search, larger ones being approached greedily. Every relation is then re-read from that order, which is re-derived for the component as a whole rather than only where relations conflict, so a relation that nothing contradicts can come back reversed too; the relations the order disagrees with are flipped rather than discarded, and both the flip and the original local score are recorded, so a reader can see how far the global correction moved. Depth propagates only along crossings, so two strands in different components are related by no evidence at all and their order is a stated tie-break, not a measurement. Supplementary Section S2 gives the sampling geometry, the score in full, the measurement that replaced an earlier two-channel rule with it, and the solver.

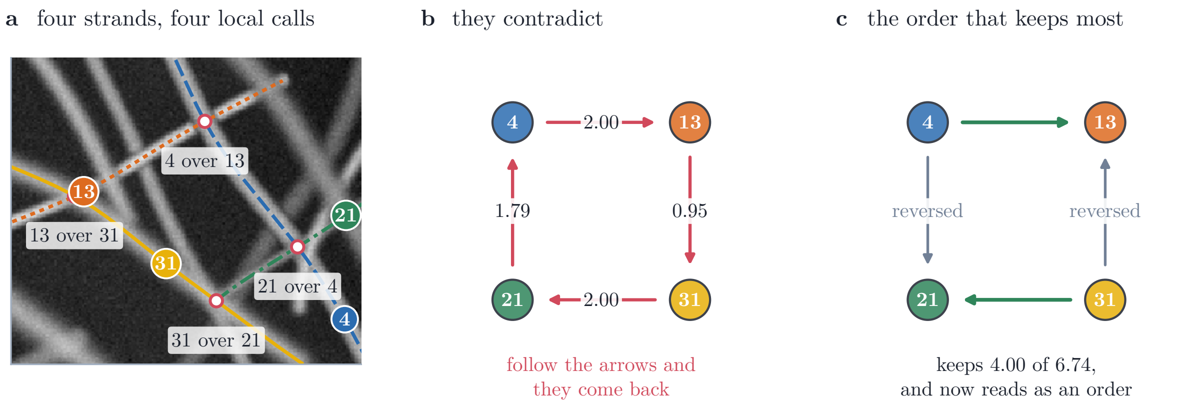


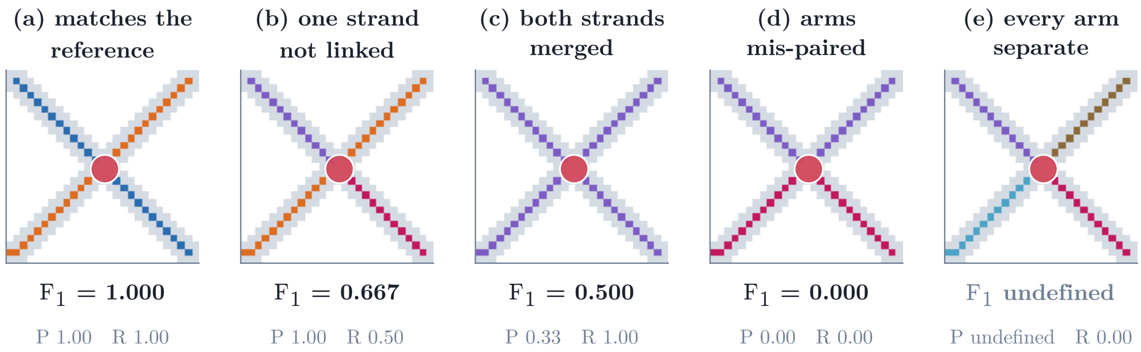
Figure 5. Why the order is solved for a whole component rather than one crossing at a time, on a contradiction the stage met. **(a)** Four strands and the four calls the local evidence made at their crossings. **(b)** Those calls as a graph: each arrow points from the strand called upper to the one called lower, and following them returns to where it started, so no stack of heights satisfies all four. **(c)** The order satisfying the most evidence by weight keeps the two heaviest calls and reverses the two lightest, which are recorded rather than discarded. Layer numbers are not shown: a strand's layer follows from its whole component, and these four sit in a scene of seven.

Layers and height. The corrected relations form a directed acyclic graph whose longest path fixes the smallest number of layers consistent with them; strands that never conflict share a layer, so the layer count follows from the ordering rather than from the number of strands. Each strand is then placed at a height that respects its order and keeps strands from interpenetrating at their measured radii, with the stack compacted so that the film is as thin as those constraints allow.

Without a greyscale image every crossing abstains, the ordering has nothing to satisfy, and the output degenerates to the tie-break: the stage requires image evidence and reports how much of it was usable. Section 3.4 evaluates it, and Supplementary Section S5 defines the measures used, which are disjoint from those above because none of them can see depth.

2.6 Metrics and statistical analysis

The primary endpoint is common-fragment pairwise F_1. The input mask is cut into fragments, each one unambiguous arm between crossings, by a rule that reads neither the reference nor any prediction, so every method compared is cut into the same pieces; each fragment then takes a reference and a predicted identity by one rule applied to both sides alike, and the score counts the pairs of fragments the prediction groups as the reference does.



Two strands cross. The scorer deletes the junction pixels (red) and keeps the four arms they separate, then scores the 6 pairs of arms: a pair is right when the grouping joins it, or leaves it apart, as the reference does. The reference is the pairing in (a); P and R are the precision and recall of those 6 pair decisions.

Figure 6. What the endpoint scores, on the smallest case that has a crossing. Two strands cross; the scorer removes a neighbourhood of the junction (red) and keeps the four arms it separates, then asks of each of the six pairs of arms whether the grouping joins it, or leaves it apart, as the reference does. **(a)** is the reference pairing. **(b)** leaves one strand unlinked, a split, which costs recall alone. **(c)** merges both strands into one instance, which costs precision alone. **(d)** pairs the arms the wrong way round and scores zero, having got no pair right. **(e)** emits every arm separately, which makes no pair at all, so precision has no denominator and F_1 is undefined rather than zero, which is one reason the adjusted Rand index and variation of information are reported beside it. Every value is computed by the scorer of Supplementary Section S3, not worked out by hand.

Supplementary Section S3 gives the construction and the assignment rule in full. Because junction pixels are deleted before fragments are labelled, the measure cannot see whether an emitted crossing pixel is given one owner or two, only whether the arms meeting there are grouped as the reference groups them; Section 3.3 separates the two.

No one measure is quoted alone, because each is blind to something another sees: pairwise F_1 scores relationships between pieces and is quadratic in fragmentation, while object detection scores existence and cannot see an arm swap at a crossing, which it rates 1.000 where pairwise scores 0.000. The adjusted Rand index and variation of information [40, 41] are therefore quoted beside the pairwise score, with detection F_1, reference-instance recovery and Crossing Fidelity. The input masks themselves are characterised by a tolerant recall, precision and Jaccard index, called completeness, correctness and quality in the thin-structure literature, matched within a tolerance rather than pixel for pixel, because two skeletonisations of one strand seldom fall on the same pixels. Supplementary Section S3 defines each in full, with its equation, what it rewards and what it cannot see.

Supplementary Section S3 states the interval construction, the resampling unit and the cases where no inference is drawn.

3 Results

3.1 Reconstruction on synthetic images

PLECTA is scored first where the answer is known exactly: the held-out synthetic scenes, under the measures of Section 2.6 and against the overlap-aware reference the generator recorded when it drew them. Table 1 reports every measure overall and by density stratum; the paragraphs below take the headline score, how it moves with density, and what that density label does and does not mean.

Table 1. Held-out reconstruction scores overall and by target areal coverage. VI denotes variation of information, for which lower is better; higher is better for the other metrics.

Group	n	F_1	Precision	Recall	Recovery	ARI	VI split	VI merge	VI total
All	128	0.854	0.868	0.843	0.799	0.851	0.256	0.216	0.473
20%	32	0.911	0.928	0.896	0.918	0.907	0.137	0.093	0.230
30%	32	0.874	0.892	0.858	0.828	0.870	0.211	0.161	0.372
40%	32	0.858	0.864	0.853	0.796	0.855	0.242	0.224	0.466
60%	32	0.775	0.787	0.765	0.656	0.772	0.436	0.388	0.823

Table 2. Bundle width against centreline length density, 36 development scenes. Within each row the three bundle widths share one byte-identical axis mask and one reference, so the spread in areal coverage carries no score difference. Length density is centreline pixels per image pixel.

Centreline length density	n	Areal coverage	Bundle width	F_1
0.0193	12	0.118–0.309	6, 11, 16 px	0.920
0.0372	12	0.218–0.505	6, 11, 16 px	0.878
0.0548	12	0.305–0.644	6, 11, 16 px	0.836

Mean common-fragment pairwise F_1 over the 128 held-out scenes was 0.854 (density-stratified 95% bootstrap CI [0.845, 0.864]); Table 1 gives precision, recall, reference-instance recovery, adjusted Rand index and both variation-of-information components, overall and by density. These scenes used unseen seeds from the same procedural generator as development data, so this is same-generator generalisation and not distribution transfer (Section 4.1).

Performance decreased with scene density: mean F_1 fell from 0.911 [0.888, 0.933] at 20% coverage to 0.775 [0.760, 0.790] at 60%, monotonically through the intermediate strata. The same decline appears on the paired clean-against-degraded set below, where the clean and the degraded curves separate at low coverage and meet at 50–60 %: degradation costs least exactly where the score is lowest, because there the tangles rather than the gaps set the difficulty. Per-scene runtime had median 1.12 s, ranging to 34.06 s on the densest scenes; Section 3.3 measures cost properly, where the transferable quantity is the scaling exponent rather than any absolute time.

Areal coverage labels these strata but does not cause the difficulty they report. Table 2 separates the two: rendering one geometry at three bundle widths leaves the input axis mask and the overlap-aware reference byte-identical while areal coverage varies by up to a factor of 2.49, and the within-geometry F_1 range across those renderings was 0.0000. Centreline length density does move the score, mean F_1 falling from 0.920 to 0.836, and the coverage bands of the three densities overlap, so a coverage stratum should be read as a proxy for the length density accompanying it under this generator.

How much of that score the input mask fixes, rather than the grouping, is separable. On the separate 50-scene controlled set, mean F_1 was 0.822 from degraded axes and 0.844 from the corresponding clean axes. The paired clean-minus-degraded change was 0.0224 (95% CI [0.0113, 0.0333]). Modelled gaps and raster irregularities therefore caused a small resolved loss under this generator.

Figure 7 shows what the score corresponds to scene by scene, taking at three coverage levels the scene whose F_1 is closest to the median, so the rows are typical rather than selected.

3.2 Development ablations

Everything here is measured on the development scenes and none of it reuses the held-out set.

What each part is worth. Table 3 varies one thing at a time. Two rows change what a link costs, dropping the junction chord-turn term or the curvature term; two change the graph

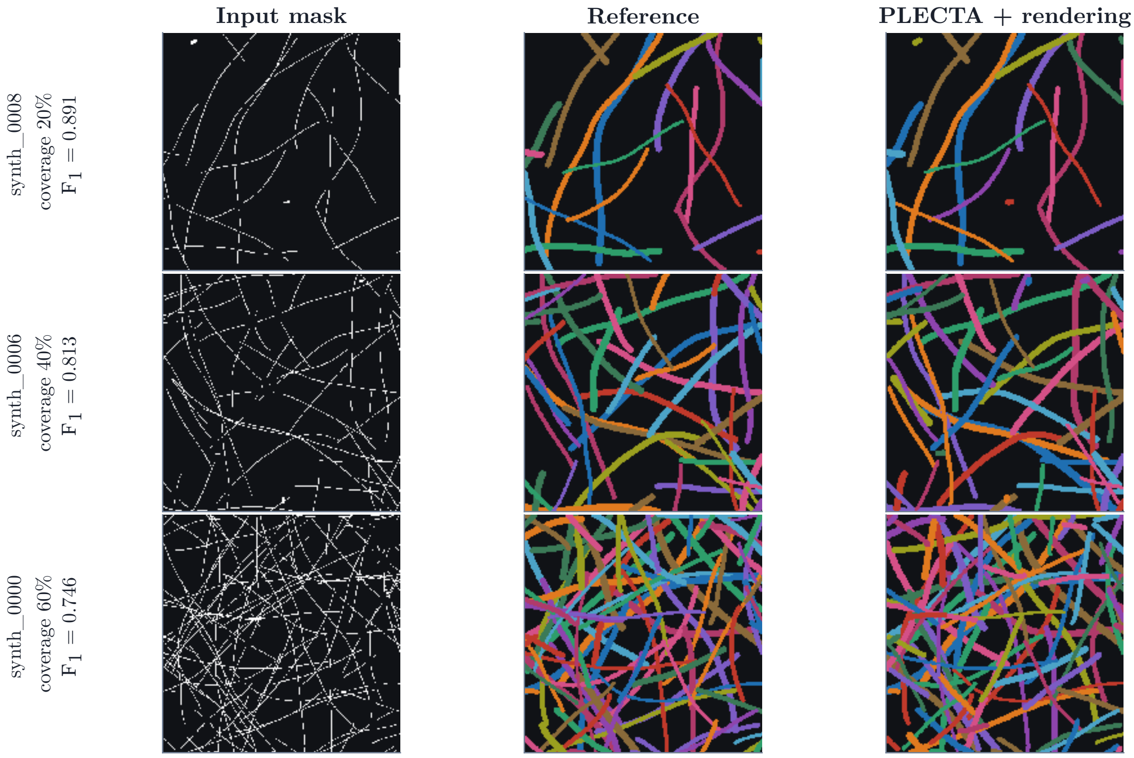


Figure 7. Reconstruction at three coverage levels, each the scene closest to the median F_1 of its stratum, shown as a horizontal window of the scene. Columns: the degraded input mask, the overlap-aware reference, and PLECTA's strands at their fitted widths. The score beside each row is the grouping's, not the rendering's. Strand colours are arbitrary and not matched between columns.

the linker reads, leaving neighbouring junction clusters unconsolidated or arms too short to merge the crossings they join; two change the iteration, running a single round or removing the schedule that relaxes the gates; and one removes gap bridging, so identity never crosses a break in the mask. The three largest losses are the junction chord-turn term, gap bridging and junction-cluster consolidation.

Two rows are controls rather than components. Removing the joint solve leaves PLECTA otherwise intact and pairs each node's arms cheapest edge first, rather than by whichever complete pairing of that node is cheapest overall; a cheap edge taken early can then strand two arms that would have paired well together, which is why that arm fails by division rather than by merging. It also runs at parameters that were selected with the exact matcher in the loop, a bias in the exact matcher's favour. The last row replaces the linker outright, skeletonising the same mask and greedily pairing the branch ends that turn least inside fixed distance and angle gates, which is the pairing principle shared by SIFNE, GraFT's path cover and Wegmayr et al. [8, 25, 26].

3.3 Published methods on identical masks

Four published methods are scored beside PLECTA on identical masks, through the same conversion and metric code, over 50 scenes generated after every configuration was fixed. What is compared is narrow: the overlap-aware grouping of a binary axis mask of dense, crossing strands, at the coverages and tangling of the scenes described in Section 2.2. Each of these methods was built for a different setting, and how it performs in that setting is not what these numbers measure. DNAi [7], GraFT [8] and SIFNE [25] are run from their own released sources. For the fourth, Basu et al. [30], no implementation was found, and it is reimplemented here.

Table 3. Component ablations and the two controls, on development scenes. The three blocks use different scene sets, so n is given per row and each ΔF_1 is against PLECTA on that row's own scenes; the column is not comparable across rows of different n.

Variant	n	F_1	ΔF_1	ARI
Fixed configuration	84	0.867	+0.000	0.864
No joint junction solve	30	0.778	-0.080	0.774
No gap bridging	84	0.728	-0.139	0.722
Single round	84	0.847	-0.020	0.844
No chain-extended frames	84	0.846	-0.021	0.842
No round schedule	84	0.863	-0.004	0.860
No curvature term	84	0.863	-0.005	0.859
No short-connector node merging	84	0.845	-0.022	0.841
No junction-cluster consolidation	84	0.804	-0.063	0.799
No junction chord-turn term	84	0.681	-0.186	0.675
Whole linker replaced	50	0.676	-0.145	0.670

Table 4. PLECTA against four published methods, identical degraded masks and one scorer, ordered by F_1. $n = 50$ scenes, GraFT over the 47 it completed. Variation of information in bits, lower better; higher better for the rest.

* our reimplementation; no implementation of that paper was found. Supplementary Section S4 details it, and the timings that column cannot support.

Measure	PLECTA	SIFNE	Basu*	GraFT	DNAi
Common-fragment F_1	0.822	0.711	0.543	0.490	0.343
Precision	0.831	0.767	0.663	0.666	0.462
Recall	0.814	0.666	0.466	0.400	0.284
Reference-instance recovery	0.765	0.606	0.385	0.304	0.252
Adjusted Rand index	0.817	0.704	0.533	0.479	0.331
Variation of information ↓	0.573	0.928	1.368	1.558	2.107
Crossing Fidelity	0.809	0.774	0.761	0.683	0.581
Median s per scene, 20/60%	0.14–4.1	3.42–5.7	2.56–54.2	2.44–29.7	0.04–0.6

Running someone else's method on data it was not built for takes decisions, and each of those decisions is stated here.

How each was run. Each released method has a front end that precedes the stage assigning identity, and in each case the mask is injected where that stage begins, so grouping is not confounded with segmentation and every stage that assigns identity is their own code. Each was also run at a configuration selected on the same 84 development scenes PLECTA was configured on rather than at its own defaults. Every one of these adaptations favours the comparator, and by a wide margin: run end to end on the greyscale, GraFT scores 0.100 against 0.490 injected, and the re-selected configurations are worth 0.195 to 0.343 for DNAi and 0.550 to 0.711 for SIFNE. Both are reported. Supplementary Section S4 gives each injection point, each selection, what was not attempted, and how each method fails.

Favoured in all those ways, the four order as Table 4 shows. SIFNE is the strongest of them, so the margin rests on a method configured for this data rather than on a default that suits it badly. Two entries read against that ordering. The first is DNAi, whose deficit is a limitation of domain and density rather than a general ranking: it is built for DNA-fibre micrographs, which are sparse, and where the network is sparse and the axis clean it recovers 0.660 at 20% coverage. The second is Basu et al. [30], the closest structural relative PLECTA's junction matching has in that it too optimises over a neighbourhood and prices joins on curvature. It stands second only to PLECTA on Crossing Fidelity while its pairwise F_1 sits far below both released methods,

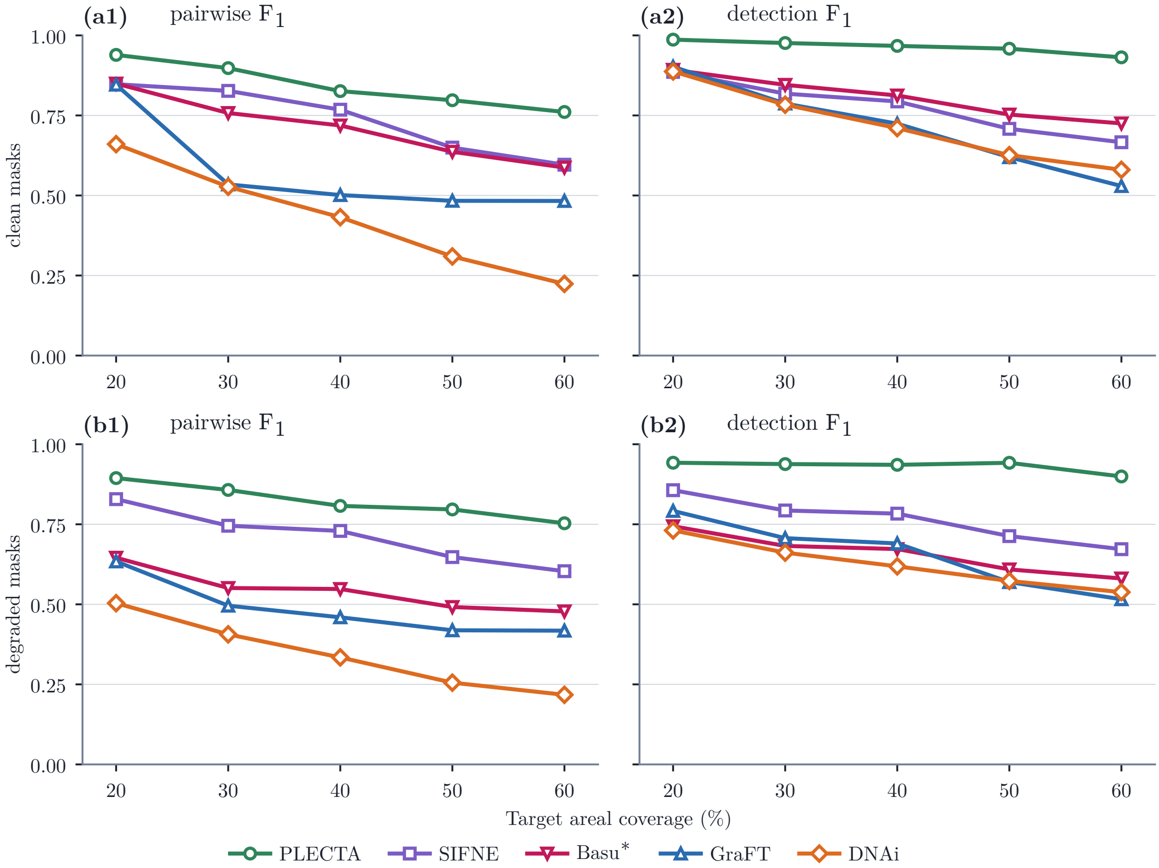


Figure 8. Five methods on identical masks, by coverage stratum: **(a)** the undegraded axis, **(b)** the degraded one; columns are pairwise F_1, which counts pairs of fragments, and detection F_1 at IoU 0.1 with a 2 px tolerance, which counts objects (Section 2.6). The two disagree where a method fragments, since only the first is quadratic in fragmentation. Crossing Fidelity is in Table 4. Means of ten scenes, no interval claimed. * our reimplementation; none was found.

so the curvature-priced matching does real work at the junctions and what costs it is everything around that decision. That is why the priced refusal of Section 2.4 is claimed as part of a combination rather than on its own. Supplementary Section S4 gives both in full.

Each method on its own ground. That ordering is drawn from one generator, so it was tested in the regimes the two released methods were built for. On GraFT's own generator, run unmodified at its published settings over 40 scenes in 8 density strata, PLECTA leads at every density and nothing crosses over, including inside the range GraFT itself published, and the field-standard measures order the three identically. SIFNE released no generator, so the sparse cobweb topology its paper validates against was rebuilt here from that description; on that reconstruction the two are level where the paper works, 0.966 against 0.966, and part only slightly at the denser end. That places the difference in how tangled the crossings are rather than in how much strand the frame holds, though a topology we rebuilt is weaker evidence than a generator its authors released. Supplementary Section S4 gives both density series in full.

What the margin is made of. An ordering says which method wins, not where. The Crossing Fidelity row of Table 4 says where: pooled over the 3638 reference crossings of the degraded set, PLECTA partitions 0.809 of them exactly, ahead of every comparator, and clean axes order them the same way. That measure is far less discriminative than the pairwise score, spanning 0.259 between best and worst method at 60 % coverage where pairwise F_1 spans 0.536, so it is used to say what was solved and not to rank. Every method is poor at the busiest nodes: above degree four PLECTA resolves 0.414 and no comparator does better, while DNAi collapses

Table 5. The depth stage on 12 depth-resolved synthetic scenes. Chance is 0.5 for every accuracy shown. Exploratory, six scenes per coverage, plain means, no inference.

	Given 2-D geometry		Reconstructed 2-D	
Areal coverage	20 %	40 %	20 %	40 %
Projected crossings per scene	28	150	28	150
Crossings recovered	1.000	1.000	0.857	0.746
Order accuracy, all crossings	0.815	0.744	0.729	0.556
Order accuracy, decided only	0.890	0.885	0.961	0.906
Abstained	0.085	0.161	0.117	0.179
Depth order, crossing pairs	0.863	0.856	0.915	0.852
Depth order, all pairs	0.671	0.708	0.703	0.715
Instance pairs sharing a component	0.392	0.842	0.361	0.746
Layer agreement	0.656	0.418	0.702	0.382

to 0.016, which is what its degree-four cap predicts. Supplementary Section S4 gives the paired margins and the all-unpaired baseline the rate is read against.

Cost. Accuracy is not the only axis on which these methods differ. Per-scene cost was measured warm, single-threaded and pinned to one core, timing each method's own computation only (Table 4; 3 GraFT scenes exceeded 600 s and are censored). These are independently optimised codebases, so absolute times measure implementation as much as algorithm; the transferable quantity is scaling, and the log–log exponents against foreground pixels are 2.11, 2.65 and at least 2.01. PLECTA has the largest uncensored exponent; GraFT's is a lower bound, because the scenes its fit lost to censoring are the slowest ones, and only GraFT is heavy-tailed.

Representing a crossing, and where it over-marks. Resolving a crossing and representing one are different things, and only the first is what the scores above measure: flattening PLECTA's emitted layers to a single label image leaves its F_1 unchanged to four decimal places. The representation is verifiable on its own, and the result is mixed. PLECTA recovers 0.463 of the pixels the reference gives to more than one strand, less than half, though the largest such recall here, against 0.144 for GraFT and 0.128 for DNAi, while the greedy baseline, a hard partition, scores zero by construction. It does so at a precision of 0.134, marking 3.6–4.0 times more shared pixels than the reference contains, because a chain is painted with every pixel of every junction cluster it touches. That is a property of the rendering rather than of the grouping, and one the F_1 neither rewards nor penalises. Supplementary Section S3 gives the shares emitted against the shares the reference carries, by coverage.

3.4 Depth decomposition

The stages above reconstruct strands in projection. A crossing pixel is assigned to both strands that own it, but nothing so far says which of the two lies on top. The optional depth stage answers that from the greyscale alone, the continuity of each strand's own brightness across the crossing weighed against the noise measured there, then imposes one globally consistent vertical order and places each strand at a height at which nothing interpenetrates. It changes no 2-D strand.

It is evaluated on 12 depth-resolved synthetic scenes at two areal coverages, under the measures of Supplementary Section S5, in two input conditions. Given the reference 2-D geometry, the depth decision is measured alone; given only the binary axis mask, the same stage runs on PLECTA's own reconstruction and carries its error. Table 5 reports both.

The decomposition is the result. Order accuracy over decided crossings spans 0.077 (0.885–0.961) across both conditions and both densities, while accuracy over all crossings spans more

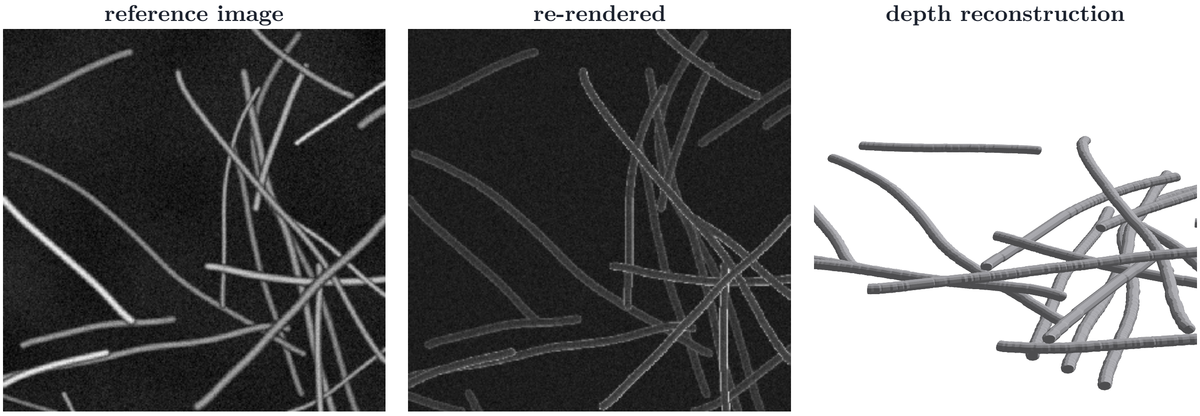


Figure 9. The depth stage on one synthetic scene, reconstructed from the binary axis mask. The image the scene was rendered from, that scene re-rendered from the reconstruction through a forward SEM model, and the film the reconstruction implies. The re-rendered panel is a consistency check on depths and radii, not a scored quantity: the difference Table 5 reports between input conditions is a difference in which crossings were *found*, which a rendered image cannot show. The table carries that measurement.

than three times that, 0.259 (0.556–0.815), and it falls by the amount the crossing recovery rate falls, 1.000 to 0.746: at the denser coverage the reconstructed condition's 0.556 over all crossings, divided by the given condition's 0.744, reproduces its crossing recovery rate 0.746 to within two thousandths. The decided figures are only meaningful beside the abstention row above them: what the stage declines there are the hard crossings, so a higher decided accuracy is a stage answering less often, not a stage deciding better. Once a crossing is found, the decision is made at a broadly similar accuracy whether the geometry was given or reconstructed, and whether the scene carries the sparser crossing load or roughly five times it, though not at the same accuracy: the widest gap between the input conditions is 0.961 reconstructed against 0.890 given, at the sparser coverage and in favour of the reconstruction, which scores only the crossings it matched and abstains on more of them. What limits the depth result is the 2-D stage not recovering every crossing, not the ordering of the crossings it does recover.

The global stack is weaker than the local decision, and the gap is a property of the evidence rather than of the decision. Depth order is recovered at 0.915 over pairs that cross but only 0.703 over all pairs, because depth propagates solely along crossings: two strands in different components of the crossing graph are related by no evidence whatever, and their order is a stated tie-break. That is measurable, not merely arguable. Only 0.361 to 0.842 of strand pairs lie within one component, and if the pairs that do are ordered at the crossing-pair rate while the rest are a coin flip, the all-pairs figure follows arithmetically: at 20 % coverage that predicts 0.650 against 0.703 measured. The arithmetic is an approximation in both directions rather than an account that holds at one end and fails at the other: it under-predicts at the sparser coverage, where the pairs in different components are evidently not quite a coin flip, and over-predicts at the denser end, where the measured order falls below the prediction by up to 0.092, consistent with error accumulating along the long transitive chains that large components create. Depth claims are therefore scoped to the connected component, and a globally ordered film would need evidence that does not travel through crossings, such as defocus, or a second view, rather than a smoothness assumption, which these scenes would not support: their heights are drawn independently, so proximity carries no depth information by construction.

Layer agreement follows the same pattern, falling from 0.702 to 0.382 as coverage doubles while the recovered layer count stays close to the reference: the ordering is locally right, so the number of layers is about right, while small order errors move strands across layer boundaries.

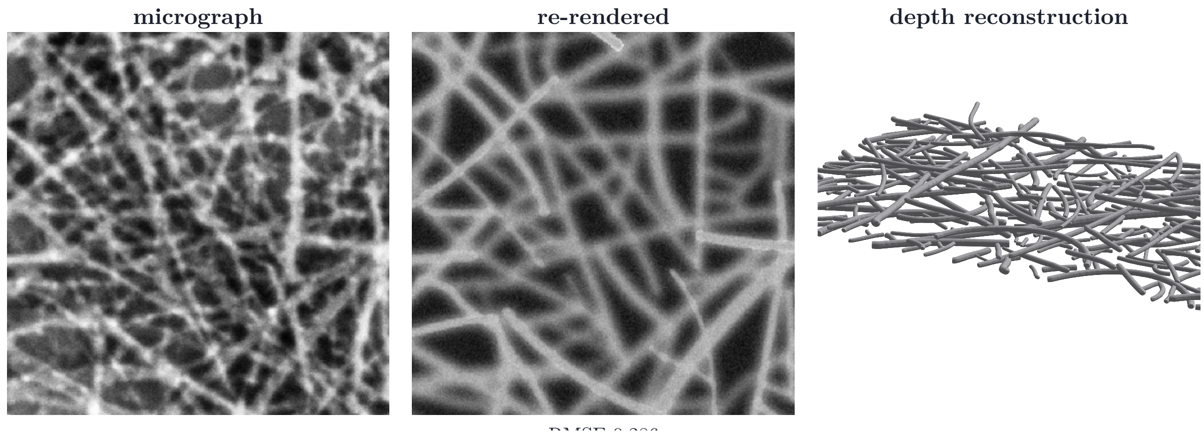


Figure 10. The same comparison on a real CNT SEM field, reconstructed from the axis of annotator A's annotation. **Nothing here is scored.** No depth ground truth exists for real SEM, because the annotation records which pixels belong to which strand, not which strand lies on top, so this shows the stage runs end to end on real data and re-renders to something recognisable, not that the depths are right. The forward model's appearance parameters are fitted to this micrograph, so the residual quoted beneath the panel is in-sample and is a sanity check rather than a measure of reconstruction quality.

3.5 Proof of concept on real SEM

The claim of this section is one sentence: PLECTA works on real imaging when a second technique supplies the mask, and Pipeline A is such a technique. Nothing here is a measurement of Pipeline A, which carries its own evaluation [34].

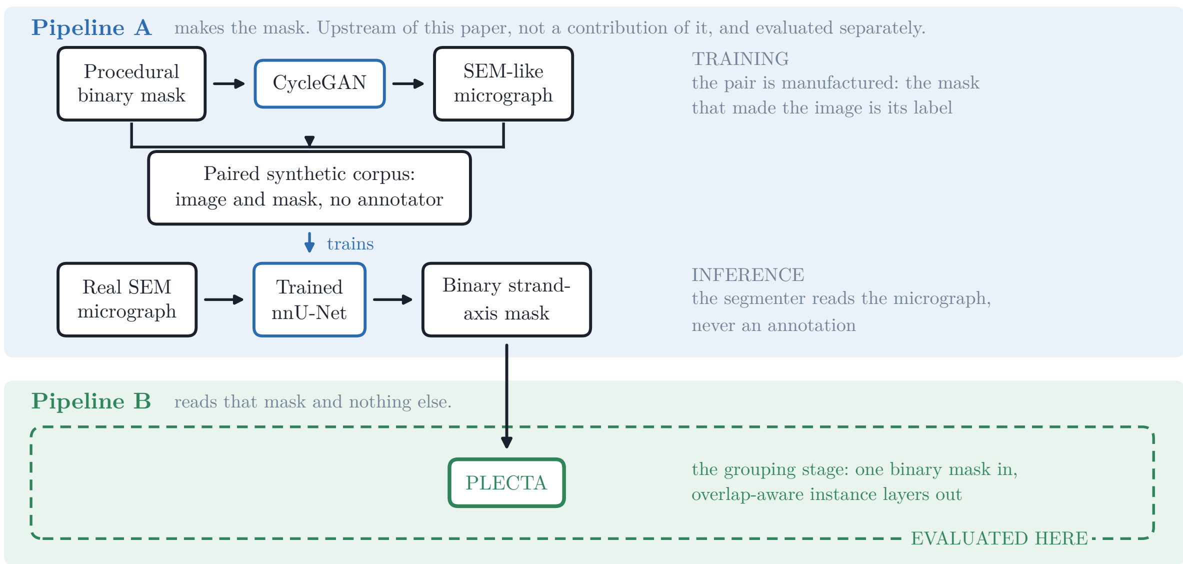


Figure 11. Where the mask comes from. Pipeline A trains a segmenter without an annotator: a generator draws a binary mask, a CycleGAN turns it into an SEM-like micrograph, and the mask that made the image is its label, so the pairs an nnU-Net trains on are manufactured rather than drawn by hand. At inference the trained network segments a real micrograph into a binary strand-axis mask. That mask is the whole interface between the two pipelines. Pipeline A is not a contribution of this paper and is not evaluated in it; it carries its own evaluation [33, 34]. Only the dashed box is measured here.

Pipeline A (Figure 11) runs procedural mask → CycleGAN → paired synthetic corpus → nnU-Net → binary strand-axis mask: a generator draws a mask, a CycleGAN translates it into an SEM-like image so that image and mask are paired without an annotator, an nnU-Net trained on

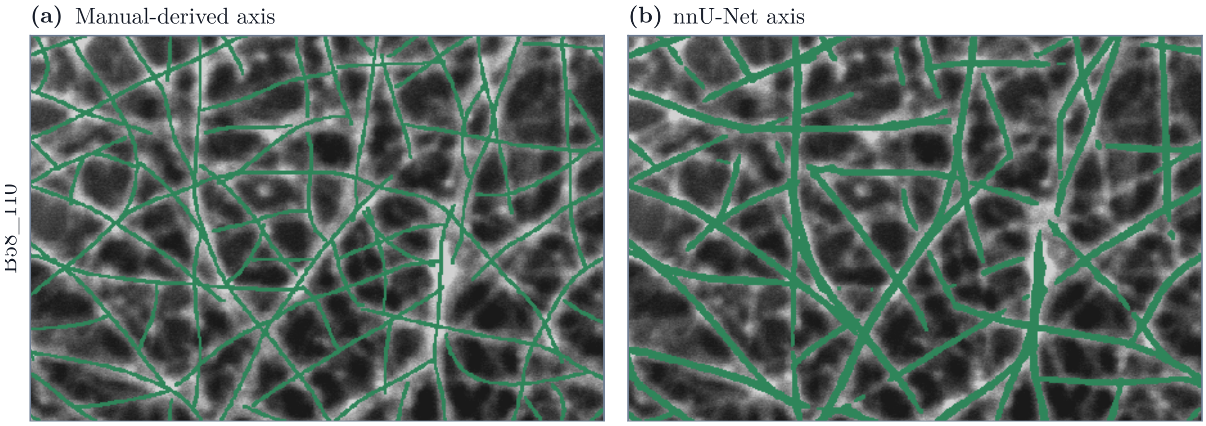


Figure 12. The two input axes of field B58_110, drawn over the micrograph they were read from: **(a)** the skeleton of annotator A's annotation and **(b)** the output of Pipeline A. These are the binary masks themselves, before any reconstruction, so there is nothing to tell apart within a panel and one colour is sufficient; the comparison the figure makes is between them. Everything separating the two conditions of Table 6 is visible here: the same strands, read a second time by a network rather than by a person, arrive thicker, broken into more pieces, and displaced by a pixel or two almost everywhere. The axis is dilated by one pixel for legibility at this size, which changes no geometry. B58_300 behaves the same way, as its mask-quality triple in the text records.

Table 6. Reconstruction from the manual-derived and the nnU-Net axis in three CNT SEM fields, under the measures of Section 2.6. *CF* is Crossing Fidelity and *Unpaired* the fidelity of the all-unpaired baseline, the do-nothing policy that leaves every arm unpaired on that field. *Cross.* is the number of reference crossings scored. Detection F_1 is at IoU 0.1 with the 2 px placement tolerance. Exploratory, 3 fields, no inference.

		Grouping			Objects and crossings			
Field	Input axis	F_1	Join F_1	Decisions	Det. F_1	CF	Unpaired	Cross.
B58_100	Manual-derived	0.887	0.929	13 234	0.941	0.922	0.013	371
B58_100	U-Net	0.457	0.559	17 282	0.633	0.617	0.102	384
B58_110	Manual-derived	0.837	0.911	24 870	0.930	0.920	0.023	599
B58_110	U-Net	0.452	0.593	16 134	0.602	0.614	0.048	290
B58_300	Manual-derived	0.947	0.959	28 224	0.959	0.956	0.011	611
B58_300	U-Net	0.466	0.615	16 729	0.577	0.617	0.048	311

those pairs segments a real micrograph, and its output is a binary axis mask [33, 34]. Pipeline B is PLECTA, and it is the only stage evaluated here. Three manually annotated CNT-sheet fields separate the two: PLECTA is run once on the skeleton of annotator A's annotation and once on the nnU-Net axis, with annotator A's annotation as the reference in both cases, so the two conditions differ only in the mask. The first is an oracle-like input condition rather than an end-to-end result, asking what the grouping can do when the axis evidence is curated; only the second says what an automatic mask allows. Figures 13 and 12 show both on one field, at width and as the centrelines the scores are computed on.

What the three fields say. From the manual-derived axis PLECTA reached F_1=0.887, 0.837 and 0.947, and resolved 0.922, 0.920 and 0.956 of the reference crossings exactly against an all-unpaired baseline of 0.011 to 0.023. From the nnU-Net axis the same predictions read 0.452 and 0.466 on B58_110 and B58_300, with Crossing Fidelity 0.614 and 0.617 and join F_1 0.593 and 0.615. Those two fields agree with each other to within 0.03 on every measure; 3 fields still cannot estimate a population, and the inferential unit is the field: the 2566 reference crossings the crossing measures rest on are pooled over six condition-specific evaluations of these three fields, a pooled count and not a sample size.

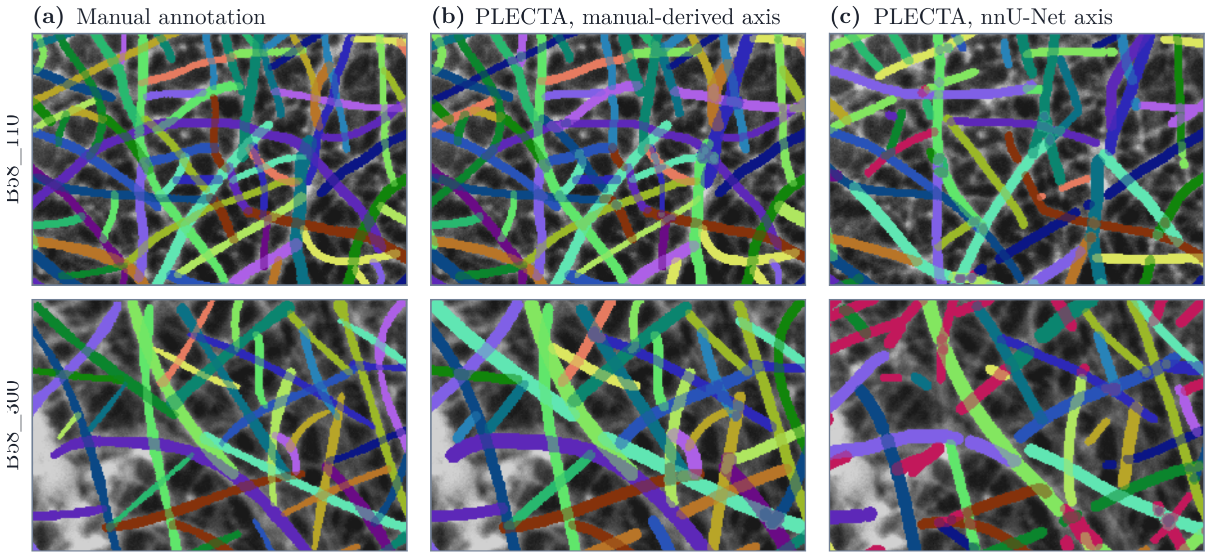


Figure 13. Reconstructed bundles on two annotated SEM fields, one per row. **(a)** annotator A's annotation; **(b)** PLECTA from the manual-derived axis; **(c)** PLECTA from the nnU-Net axis of Figure 12b, the same method and parameters, differing only in the mask. All three are drawn at width, the reconstructions through the optional layer of Section 2.5.1, which runs after the grouping is fixed and moves no pixel between strands. Within a row, a predicted strand takes the colour of the annotated strand it matched; crimson matched none. Each crop is the window whose local pairwise F_1 is closest to the whole-field value on both axes.

Most of the distance between the two conditions is the mask, and the method fails by conservatism rather than by fusing. The axis each field hands PLECTA is characterised, before any reconstruction is opened, by the tolerant completeness/correctness/quality triple: at 2 px, quality is 0.504, 0.557 and 0.507, over completeness 0.671, 0.651 and 0.619. The tolerance is not a formality: quality falls to 0.299, 0.375 and 0.325 at 1 px, because the annotation and the nnU-Net output are two independent readings of the same strands and disagree in placement almost everywhere. That displacement is also why detection F_1 carries a placement tolerance, and the control that makes it a correction rather than an inflation is here: on the manual-derived axes, where the input mask is a skeleton of the annotation itself, the tolerance moves detection F_1 by 0.000, against 0.179 rising to 0.602 on B58_110's nnU-Net axis. Because the pairwise score is quadratic in fragments, a strand the nnU-Net mask breaks into several pieces carries every one of its pairs. Join F_1 removes that weighting by scoring one decision per pair of fragment ends the mask brings within reach, rather than every pair of fragments the strand has; it is defined in Supplementary Section S3 with the rest, and quoted only here, where a difference in fragmentation is what is being read. It reports 24870 decisions on B58_110's manual axis against 16134 on its nnU-Net one. It narrows the gap without closing it, and its decomposition says what the rest is: a join precision of 0.681 against a recall of 0.525 is a method declining joins rather than fusing strands, which is what panel (c) of Figure 13 shows.

4 Discussion

PLECTA addresses a narrow problem: assigning fragments of a binary strand-axis mask to overlapping projected instances. Explicit geometry recovers substantial identity there with no instance-labelled training data and no image intensity, and what recovers it is the combination of local and accumulated evidence, neither of which suffices alone: a junction is resolved by mutual compatibility rather than by the cheapest available continuation, and the evidence for that judgement is re-read along the chains as they assemble. That evidence is directional throughout, which fixes the condition the image must meet: a tangent measured on one side of a crossing has to predict the arm leaving the other, which bending stiffness produces most directly and tension, confinement or bundling can produce as well. The ablations are consistent with that reading, the two largest losses being the junction chord-turn term and gap matching, both purely geometric. Where a strand is flexible on the scale of a junction the premise fails rather than degrades: there is then nothing local to match, and the correct response is to acquire evidence the projection has lost rather than to weaken the gates. The ablations locate that effect without proving each term independently necessary: the two large losses correspond to the two kinds of discontinuity, the junction and the missing mask segment, while accumulated context is worth distinctly less than either.

Difficulty follows the density of centrelines rather than the amount of ink, so what costs the score is the number of locally plausible alternatives at a crossing. The rest of the sensitivity belongs to the upstream segmenter, which fixes how clean a skeleton PLECTA is given. PLECTA can bridge a short omission, but it cannot infer a strand absent from the mask, nor reliably reject a false axis introduced upstream. Supplying identical masks isolates grouping in an algorithmic comparison; it does not establish end-to-end SEM performance, which is why the three-field case is a demonstration of the joined pipelines rather than a measurement of them. Ribbon rendering, being downstream of grouping, buys connectivity rather than identity, because the core paints arms and junction clusters but not the gap links that carry identity across a break; width and brightness can be reported for characterised instances, but their physical accuracy requires separate references.

4.1 Evidence boundary

The principal limitation is distributional, and it is narrower than a one-generator study. The held-out scenes use independent seeds but the same procedural generator family used for development, so their confidence intervals quantify sampling variation within that family and not transfer to another simulator, specimen, microscope, or segmentation pipeline. The *ranking*, however, does not rest on that family: GraFT's own generator was run unmodified at its published settings over 40 scenes in 8 density strata, and its bundled frame separately, and the ordering held on both (Section 3.3). What is untested is transfer of the held-out *level* to a third generator, not of the comparison. The ablation and rendering studies are development-only and remain secondary to the fixed held-out result.

Within that family, one generator setting is worth singling out because the method's grouping core is an angle-continuation argument and a reader may reasonably suspect the result of resting on it. The generator draws each centreline as a correlated-curvature random walk whose bending stiffness is one parameter, set to 0.035 rad per step because CNT bundles are stiff. Sweeping it from 50 % to 150 % of that value, which moves the median radius of curvature from 975 to 308 px in a 512 px frame, changes nothing that can be distinguished from noise: over 5 settings at two coverage targets and both mask conditions, 10 scenes each, the largest spread of level means is 0.048 against a smallest scene-to-scene standard deviation of 0.050. The headline numbers do not depend on this parameter across that band. The wavelength over which a bend is spread was swept separately, from 10 to 40 px about a default of 20, with the same result in 3 of 4 series; the fourth exceeds its own noise only marginally (0.076 against 0.052) and is non-monotonic with its minimum at the default, so it is read as noise. This is a robustness statement and not a scope bound: it locates no breaking point, and the sweep stays well inside the stiff regime the paper claims, reaching a 95th-percentile local bend of only 9.2° over 20 px.

The real SEM study is deliberately a case study with 3 fields, not a population validation. Manual-derived masks are an oracle-like input condition: they test grouping when the axis evidence is curated, not automatic SEM-to-instance performance. Three fields cannot estimate generalisation, however closely any two of them agree. The two mask sources also generate different fragment universes, so their F_1 values describe performance under each input condition rather than comparing identical objects. More independent fields, sealed before configuration, are still required.

Independent re-annotation is no longer outstanding, and the result reframes the limitation above rather than softening it. Annotator B independently annotated 3 of these fields. Scored against the annotator whose skeleton it was given, A against A and B against B, PLECTA reaches 0.786–0.947; scored across, A against B and B against A, it reaches 0.370–0.510. That collapse is not the method degrading. A and B agree with each other at 0.361–0.535 by the same measure with no method in the loop, so the cross-annotator scores sit at roughly the ceiling human agreement allows. The disagreement is systematic and not noise: B exports 1.85–2.83 times as many objects as A, shorter, with 8–17 times as many components in the per-instance-skeleton union, and the directional variation of information is split-dominant one way and merge-dominant the other. That is two different answers to what an instance *is*, a whole physical strand or a locally visible segment, not one annotator being wrong. For the same reason the automatic-mask condition, at 0.383–0.466 against either annotator, cannot be separated from annotator disagreement on this evidence.

This is also the strongest available argument for evaluating on a generator at all. Instance identity through an occlusion is precisely the quantity manual annotation cannot fix: on real fields the reference is a reading, and two competent readings of the same micrograph differ by more than most of the method gaps this paper reports. A procedural scene is the only condition in which the answer is defined rather than adjudicated, because identity is assigned before rendering rather than recovered after it. The synthetic evaluation is therefore not a convenience standing in for real ground truth; it is the only place an unambiguous instance definition exists, and the real fields are correctly read as a case study of behaviour under a stated annotation protocol.

One caveat from the comparisons must be separated from the claim it seems to threaten. The comparators lose on how a crossing is *resolved*, which the metric scores and which is the difficult decision, not on how it is *represented*, which the metric cannot see: GraFT's near-hard partition is a real limitation that costs it nothing here. Both properties are worth having, and Section 3.3 measures the representation directly, including where PLECTA over-marks it.

Two further limits concern the output itself. PLECTA recovers projected two-dimensional identity: shared pixels encode overlap but not which strand passes above another, so no three-dimensional topology follows. And the grouping metric evaluates ownership of centreline fragments, which is not a validation of rendered width; quantitative diameter claims require matched physical measurements across the relevant scale, contrast, and specimen ranges. Pairwise fragment scores also emphasise instances with many fragments, which is why recovery and the adjusted Rand index are reported alongside.

The comparisons themselves are bounded. The greedy rule is our reimplementation of a principle rather than of any particular paper, and DNAi was run outside the domain it was built for, part of its cost function reading a colour channel our masks do not have. What the results support is narrow and mechanical: pairing endpoints greedily, or committing to a pairing with no option to decline, loses identity in dense crossings. They do not support a claim that PLECTA is generally superior; that would need several released implementations run in their own regimes by their own authors. Broader validation should extend to independent real masks and other strand domains, with input fragments and scoring rules held fixed.

5 Conclusions

PLECTA reconstructs overlapping strand instances from a binary axis mask by alternating exact junction matching with gated gap matching as local arm geometry expands into chain-level evidence. Its evidence is directional throughout, so it requires tangent and curvature to persist across a crossing at the scale of its fitting support, a condition bending stiffness contributes to and rod-like strands commonly meet. That is the regime of nanotube bundles and of the cytoskeletal and synthetic-fibre networks comparable methods target, and not that of freely flexible chains, where the projection retains nothing local to match. The fixed method achieved common-fragment $F_1 = 0.854$ on 128 independently seeded held-out scenes from the same procedural generator, with performance declining as network density increased, and partitioned 0.809 of the comparison set's reference crossings exactly as the reference does. On a generator that is not ours, being GraFT's own, run unmodified at its published settings over 40 scenes in 8 density strata, it reached $F_1 = 0.916$ against 0.475 and 0.349 for the two released comparators, so the ordering does not rest on the generator that produced the held-out set. On identical masks it gained +0.145 F_1 over a greedy minimum-turning-angle continuation rule, the rule shared by several published linkers, and +0.478 and +0.339 over two released implementations run from their own source; these support solving each junction exactly rather than pairing endpoints in priority order, but the released comparisons are bounded by domain and density and do not establish a general ranking. Development ablations identify junction chord-turn evidence, gap bridging, and junction-topology consolidation as the principal contributors, and optional image-informed width reconstruction improves geometric presentation while remaining separate from the grouping claim.

The proof of concept on carbon-nanotube sheets joins the two pipelines: an upstream synthetic-data-driven segmenter prepares the mask, and PLECTA resolves it into instances, demonstrated on 3 annotated SEM fields under manual-derived and nnU-Net masks, without establishing population generalisation or width accuracy. The method therefore offers an inspectable route to projected strand identity that draws on no random number generator and, in its grouping core, no training data; the demonstrated SEM route is not training-free, since the mask it consumes comes from a trained CycleGAN and nnU-Net upstream. Broader claims require independent real fields, validated physical widths, further released implementations run in a matched comparison, and an external-generator benchmark with sealed seeds.

CRediT author contribution statement

[To do: Fill in each author's CRediT roles before submission, checked against actual contributions; keep the author names and order as listed above.]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The PLECTA source, complete fixed configuration, compact seed manifest, and aggregate machine-readable evidence are available at `https://github.com/Jorallan/PLECTA`. Generated scenes, detailed per-scene records, and development diagnostics are retained locally. **[To do: Create a tagged archival release and add its DOI before submission; do not infer or preassign a DOI.]**

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